Zhifeng Jiang

Embodied AI Researcher

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Research Interests

Embodied AI systems, *especially on vision-language-action (VLA) models, real-robot learning, first-person wearable agents, robot agents, and long-horizon manipulation.*

On-device AI systems, *especially on LLM inference acceleration (for phones), speech enhancement (for AI glasses), and tool-use agents (for personal computers).*

Experiences

Nov 2024 – Sep 2026 **Research Scientist in Central Software Institute**,
Huawei Technologies, Shenzhen, China

- Embodied AI and Robot Agents
  - Lead a 6-person team to build RobiAgent, a dual-arm robot agent prototype for studying real-robot manipulation and real-time, constraint-aware coordination of coupled robot subtasks.
  - Explore VLA-based robot manipulation through $\pi_{0.5}$ simulation reproduction, real-robot data collection, imitation learning, and deployment on SO101 arms.
- Wearable Multimodal Intelligence on AI Glasses
  - Build VisionBot, an AI-glasses-oriented agent for always-on perception and service, with demos including lost-and-found via continuous visual search and fitness coaching via first-person activity logging and feedback.
  - Develop a target speech hearing system for AI glasses by designing a stereo speech enhancement model, building a streaming inference pipeline, and deploying it through a HarmonyOS application.
- LLM/VLM Applications and Systems
  - Develop LLM/VLM applications including on-device fraud webpage detection, large-scale RAG-based question answering, and an MCP-compatible command-line agent (Hey).
  - Explore efficient inference for autoregressive language models via parallel decoding.

Sep 2019 – Jul 2024 **PhD Candidate in Computer Science and Engineering**,
The Hong Kong University of Science and Technology, Kowloon, Hong Kong

- Private Inference for LLMs
  - Protect system prompt with minimized impact on conversational quality (PromptKeeper)
- Private and Efficient Federated Training
  - Secure participant selection against adversarial servers (Lotto)
  - Enable dropout-resilient differential privacy without loss in time performance (Dordis)
  - Train asynchronously with principles for end-to-end speedup (Pisces)

Education

Sep 2019–Jul 2024 **Ph.D. in Computer Science and Engineering**,
The Hong Kong University of Science and Technology, Kowloon, Hong Kong
Dissertation: "Towards Private and Efficient Cross-Device Federated Learning"

Jul 2018 – Oct 2018 **Research Intern in Prof. Dean Tullsen's Research Group**,
University of California, San Diego, La Jolla, US

- Defense against Return-Oriented Programming with Context-Sensitive Decoding on x86-64.

Sep 2015–Jun 2019 **B.Eng. in Computer Science**,
Zhejiang University, Hangzhou, China
GPA: 3.97/4.0, Graduated with Outstanding Honor (Zhejiang Province)

Publications

Selected Publications

2025 **Zhifeng Jiang**, Zhihua Jin, Guoliang He. "PromptKeeper: Safeguarding System Prompts for LLMs" , *in the Findings of the ACL EMNLP 2025*

- 2024 **Zhifeng Jiang**, Peng Ye, Shiqi He, Wei Wang, Ruichuan Chen, Bo Li. "Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning", *in the Proc. of USENIX Security 2024*
- 2024 **Zhifeng Jiang**, Wei Wang, Ruichuan Chen. "Dordis: Efficient Federated Learning with Dropout-Resilient Differential Privacy", *in the Proc. of ACM EuroSys 2024*
- 2023 **Zhifeng Jiang**, Wei Wang, Bo Li, Qiang Yang. "Towards Efficient Synchronous Federated Training: A Survey on System Optimization Strategies", *in IEEE TBD, Volume 9, Issue 2*
- 2022 **Zhifeng Jiang**, Wei Wang, Baochun Li, Bo Li. "Pisces: Efficient Federated Learning via Guided Asynchronous Training", *in the Proc. of ACM SoCC 2022*

Other Publications

- 2026 Na Lv, Jiayi Zhang, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Quan Chen, Minyi Guo. "Enabling Client-Autonomous Training Optimizations for Efficient Federated Learning", *accepted to appear in IEEE TPDS*
- 2026 Zekai Su, Yongjun He, Jiqing Han, Jianchen Li, Yikun Jiang, **Zhifeng Jiang**, Tianhong Ding. "AS-EVNorm: Tail-Aware Extreme Value Normalization for Speaker Verification", *in IEEE Signal Processing Letters, Volume 33*
- 2026 Jiayi Zhang, Zuo Gan, Chen Chen, **Zhifeng Jiang**, Hao Wang, Yifei Zhu, Quan Chen, Minyi Guo. "Boosting Gradient-based Training Diagnosis for Efficient and Accurate Federated Learning", *accepted to appear in IEEE TMC*
- 2024 Peng Ye, **Zhifeng Jiang**, Wei Wang, Bo Li, Baochun Li. "Feature Reconstruction Attacks and Countermeasures of DNN Training in Vertical Federated Learning", *in IEEE TDSC, Volume 22, Issue 3*
- 2024 Yongkang Zhang, Haoxuan Yu, Chenxia Han, Cheng Wang, Baotong Lu, Yunzhe Li, **Zhifeng Jiang**, Yang Li, Xiaowen Chu, Huaicheng Li. "SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs", *in the Proc. of ACM PPoPP 2025*
- 2024 Na Lv, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Jiayi Zhang, Quan Chen, Minyi Guo. "FedCA: Efficient Federated Learning with Client Autonomy", *in the Proc. of ICPP 2024*
- 2021 Minchen Yu, **Zhifeng Jiang**, Hok Chun Ng, Wei Wang, Ruichuan Chen, Bo Li. "Gillis: Serving Large Neural Networks in Serverless Functions with Automatic Model Partitioning", *in the Proc. of IEEE ICDCS 2021;* ***Best Paper Runner-Up***
- 2021 **Zhifeng Jiang**, Wei Wang, Yang Liu. "FLASHE: Additively Symmetric Homomorphic Encryption for Cross-Silo Federated Learning", *in arXiv preprint (Citation: 108+)*

Honors and Awards

- 2024, 2023 Redbird Academic Excellence Award, HKUST
- 2021 Best Paper Runner-Up Award (Top 3 out of 97 accepted papers), IEEE ICDCS
- 2019 Outstanding Graduate Award (Top 1%), Zhejiang Province
- 2017 He Zhijun Scholarship (Top 10 in Dept. of CS), ZJU
- 2017 National Scholarship (Top 0.1% nationwide), Ministry of Education, China

Professional Service

- Journal Reviewer: IEEE TMC, IEEE TBD, IEEE TNNLS, IEEE TKDE, IEEE TDSC, IEEE TC.
- Conference Reviewer: ICLR 2026 Workshop RSI, ACM EuroSys 2023 (Shadow).
- AEC Member: USENIX OSDI 2022, USENIX ATC 2022, ACM SOSP 2021.

江志锋

具身智能研究员

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研究兴趣

具身智能系统, 尤其关注视觉-语言-动作（*VLA*）模型、真实机器人学习、第一人称可穿戴智能体、机器人智能体，以及长程操纵任务。

端侧AI 系统, 尤其关注面向手机的大语言模型（*LLM*）推理加速、面向*AI* 眼镜的语音增强，以及面向个人电脑的工具调用智能体。

经历

2024 年11 月– 2026 年9 月 **中央软件院研究员**,
华为技术有限公司, 深圳, 中国

- 具身智能与机器人智能体
  - 带领6 人团队构建RobiAgent，一个双臂机器人智能体原型，用于研究真实机器人操纵，以及存在实时约束的耦合机器人子任务协调。
  - 围绕基于VLA 的机器人操纵，完成$\pi_{0.5}$ 仿真复现、真实机器人数据采集、模仿学习，以及在SO101 机械臂上的部署。
- AI 眼镜上的可穿戴多模态智能
  - 构建VisionBot，一个面向AI 眼镜的常驻感知与服务智能体，已完成基于连续视觉搜索的物品寻找，以及基于第一人称活动记录和反馈的健身指导两个demo。
  - 开发面向AI 眼镜的定向拾音系统，包括设计双声道语音增强模型、构建流式推理流水线，并通过HarmonyOS 应用完成部署。
- 大语言/视觉语言模型应用与系统
  - 开发大语言/视觉语言模型应用，包括端侧涉诈网页检测、大规模检索增强问答，以及兼容MCP 的通用命令行智能体（Hey）。
  - 探索通过并行解码加速自回归语言模型的推理。

2019 年9 月– 2024 年7 月 **计算机科学与工程博士研究生**,
香港科技大学, 九龙, 中国香港

- 大语言模型推理的隐私保护
  - 在尽量降低对话质量损失的前提下保护系统提示词（PromptKeeper）。
- 隐私保护与高效联邦训练
  - 在存在恶意服务器的场景下，保护联邦学习参与方选择过程的安全性（Lotto）。
  - 在不损失时间性能的前提下，实现可抵抗客户端掉线的差分隐私联邦学习（Dordis）。
  - 基于异步训练原则实现端到端训练加速（Pisces）。

教育背景

2019 年9 月– 2024 年7 月 **计算机科学与工程博士**,
香港科技大学, 九龙, 中国香港
博士论文：“Towards Private and Efficient Cross-Device Federated Learning”

2018 年7 月– 2018 年10 月 **Prof. Dean Tullsen 研究组科研实习生**,
加州大学圣地亚哥分校, La Jolla, 美国

- 研究x86-64 平台上针对返回导向编程攻击的防御方法。

2015 年9 月– 2019 年6 月 **计算机科学与技术学士**,
浙江大学, 杭州, 中国
GPA：3.97/4.0，获浙江省优秀毕业生

论文发表

代表论文

2025 **Zhifeng Jiang**, Zhihua Jin, Guoliang He. "PromptKeeper: Safeguarding System Prompts for LLMs" , *Findings of the ACL EMNLP 2025*

2024 **Zhifeng Jiang**, Peng Ye, Shiqi He, Wei Wang, Ruichuan Chen, Bo Li. "Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning" , *USENIX Security 2024*

2024 **Zhifeng Jiang**, Wei Wang, Ruichuan Chen. "Dordis: Efficient Federated Learning with Dropout-Resilient Differential Privacy" , *ACM EuroSys 2024*

2023 **Zhifeng Jiang**, Wei Wang, Bo Li, Qiang Yang. "Towards Efficient Synchronous Federated Training: A Survey on System Optimization Strategies" , *IEEE TBD 第9 卷第2 期*

2022 **Zhifeng Jiang**, Wei Wang, Baochun Li, Bo Li. "Pisces: Efficient Federated Learning via Guided Asynchronous Training" , *ACM SoCC 2022*

其他论文

2026 Na Lv, Jiayi Zhang, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Quan Chen, Minyi Guo. "Enabling Client-Autonomous Training Optimizations for Efficient Federated Learning" , *IEEE TPDS*

2026 Zekai Su, Yongjun He, Jiqing Han, Jianchen Li, Yikun Jiang, **Zhifeng Jiang**, Tianhong Ding. "AS-EVNorm: Tail-Aware Extreme Value Normalization for Speaker Verification" , *IEEE Signal Processing Letters 第33 卷*

2026 Jiayi Zhang, Zuo Gan, Chen Chen, **Zhifeng Jiang**, Hao Wang, Yifei Zhu, Quan Chen, Minyi Guo. "Boosting Gradient-based Training Diagnosis for Efficient and Accurate Federated Learning" , *IEEE TMC*

2024 Peng Ye, **Zhifeng Jiang**, Wei Wang, Bo Li, Baochun Li. "Feature Reconstruction Attacks and Countermeasures of DNN Training in Vertical Federated Learning" , *IEEE TDSC 第22 卷第3 期*

2024 Yongkang Zhang, Haoxuan Yu, Chenxia Han, Cheng Wang, Baotong Lu, Yunzhe Li, **Zhifeng Jiang**, Yang Li, Xiaowen Chu, Huaicheng Li. "SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs" , *ACM PPoPP 2025*

2024 Na Lv, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Jiayi Zhang, Quan Chen, Minyi Guo. "FedCA: Efficient Federated Learning with Client Autonomy" , *ICPP 2024*

2021 Minchen Yu, **Zhifeng Jiang**, Hok Chun Ng, Wei Wang, Ruichuan Chen, Bo Li. "Gillis: Serving Large Neural Networks in Serverless Functions with Automatic Model Partitioning" , *IEEE ICDCS 2021*; **获最佳论文奖提名**

2021 **Zhifeng Jiang**, Wei Wang, Yang Liu. "FLASHE: Additively Symmetric Homomorphic Encryption for Cross-Silo Federated Learning" , *arXiv 预印本（引用108+）*

荣誉奖励

2024, 2023 香港科技大学红鸟学术卓越奖

2021 IEEE ICDCS 最佳论文提名奖（97 篇录用论文中前3）

2019 浙江省优秀毕业生（前1%），浙江省教育厅

2017 浙江大学何志均教育基金奖学金（计算机学院前10）

2017 国家奖学金（全国前0.1%），中华人民共和国教育部

专业服务

期刊审稿人 IEEE TMC, IEEE TBD, IEEE TNNLS, IEEE TKDE, IEEE TDSC, IEEE TC.

会议审稿人 ICLR 2026 Workshop RSI, ACM EuroSys 2023 (Shadow).

Artifact 评审委员会成员 USENIX OSDI 2022, USENIX ATC 2022, ACM SOSP 2021.